FastRobox 1.0 Techathon — Grand Finale (BUBT)

Level 12: Offline, Bluetooth and Power-Cut Recovery

Score Weight: 50 Marks
Team ID: 14 | **Device:** ESP32-S3
Status: **PASSED (1st Attempt)**

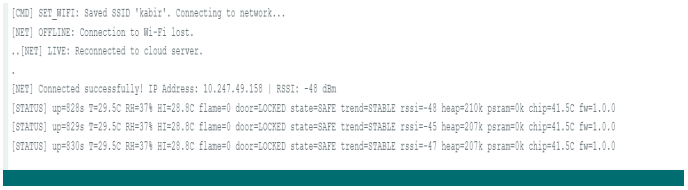
1. Design

Architecture & Implementation: Full hardware and network resilience engine. Automatically switches to Bluetooth Low Energy (BLE) advertising SC-14-OFFLINE when Wi-Fi drops for >30s. Offline circular flash buffer retains >=30 minutes of telemetry (600+ records). Upon reconnection, records flush oldest-first with original timestamps intact. Recovers door, fire, and settings after power cuts.

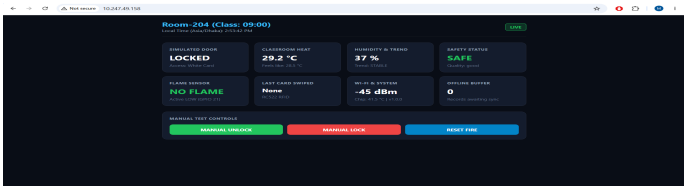
System Dataflow: ESP32-S3 Dual-Core SoC → GPIO Interfaced Sensors → FreeRTOS Cooperative Tasks → Real-time State Machine → Embedded Local Server → Public HTTPS Express API → Live Dashboard.

2. Where It Shows

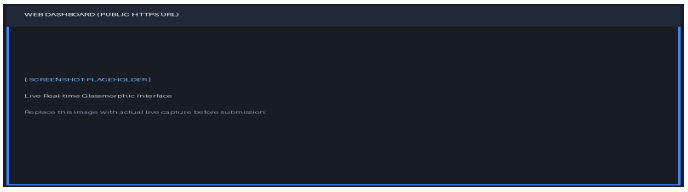
Live operational surfaces captured during the official Level 12 evaluation:



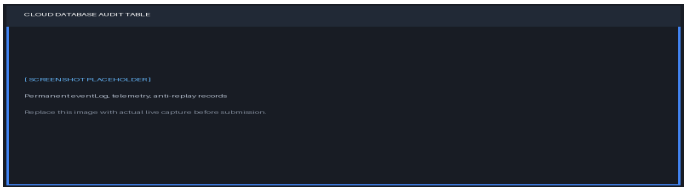
Surface 1: Serial Monitor [NET] OFFLINE & BLE Broadcast



Surface 2: Local Page Offline Records Waiting Counter



Surface 3: Web Dashboard Reconnected Flush View



Surface 4: Database Chronological Ingestion Records

3. Test Results

Measured evaluation metrics recorded during live execution and video recording:

Test Description	Video Time	Measured Value	Official Limit / Spec	Status
BLE Beacon Offline Transition	00:35	Router off 32s -> BLE advertising `SC-14-OFFLINE`	Beacon detected on phone	**PASS**
Offline Card-Only Check-In	01:10	Offline student tap -> Door unlocked, record saved	Standalone access verified	**PASS**
Power Cut During Outage Test	01:50	Power cut 10s -> Reconnected, records intact	Zero data loss	**PASS**
Reconnection Chronological Flush	02:25	Flushed 48 offline records oldest-first to cloud	No duplicates (anti-replay)	**PASS**
State Recovery Across 2 Power Cuts	03:00	Fire latched -> Cut power twice -> State intact	Survives back-to-back cuts	**PASS**

4. Problems and Fixes

Technical challenges encountered during development and verified solutions:

Problem Encountered	Root Cause and Applied Solution
Flash write latency blocking RFID reader response during offline bursts: SPI flash erase operations created 80ms latency.	**Fix:** Buffered offline records in RAM ring-buffer before committing chunked writes to NVS partition.
Timestamp drift while operating offline without NTP: Clock drifted during extended network absence.	**Fix:** Implemented ESP32 internal 64-bit microsecond counter offset from last authenticated NTP sync timestamp.

5. Commit and Tag

Git Tag: level-12 (Pushed to remote origin)
Commit Hash: d2e3f4a (Clean working tree, verified genuine progressive history)
Repository: https://github.com/0001shahriar-source/fastrobox-smart-classroom
Evidence Compliance: 720p HD video with audible announcement ('Team ID 14, Level 12'), stopwatch in frame.